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Processando config 19, seed 1, teste: Rabbit
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

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*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

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smultin_MultinomialBitsOver test:

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N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE

Number of bits = n = 8388608
Number of cells = 2^L = 274877906944
Expected number per cell = 1 / 32768
EColl = n^2 / (2k) = 128
Hashing = TRUE

Collision test
CollisionOver: density = n / k = 1 / 32768.00
Expected number of collisions = Mu = 128.00

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Results of CollisionOver test:

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POISSON approximation :
Expected number of collisions = N*Mu : 256.00
Observed number of collisions : 222
p-value of test : 0.98

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Total number of cells containing j balls

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j = 0 : 549739036894
j = 1 : 16776772
j = 2 : 222
j = 3 : 0
j = 4 : 0
j = 5 : 0

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CPU time used : 00:00:04.44

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Generator state:
16777216 bits have been read.

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*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

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snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 17.00
p-value of test : 0.39

CPU time used : 00:00:00.05

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 8.00
p-value of test : 0.12

CPU time used : 00:00:00.07

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : 0.030
p-value of test : 0.49

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 4.79
p-value of test : 0.69

Normal statistic for number of jumps : -1.31
p-value of test : 0.90

CPU time used : 00:00:00.18

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : -0.91
p-value of test : 0.82

CPU time used : 00:00:00.41

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : -0.12
p-value of test : 0.55

CPU time used : 00:00:00.19

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.018
p-value of test : 0.72

Kolmogorov-Smirnov- statistic = D- : 0.023
p-value of test : 0.56

Anderson-Darling statistic = A2 : 0.30
p-value of test : 0.94

CPU time used : 00:00:00.13

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 4.84
p-value of test : 0.18

Global longest run of 1 : 25.00
p-value of test : 0.22

CPU time used : 00:00:00.06

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 105.50
p-value of test : 0.20

CPU time used : 00:00:00.01

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 16.51
p-value of test : 0.79

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 1.39
p-value of test : 0.08

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : 0.34
p-value of test : 0.37

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : -1.22
p-value of test : 0.89

CPU time used : 00:00:00.00

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 198.91
p-value of test : 0.12

CPU time used : 00:00:00.00

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 312.55
p-value of test : 0.37

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 465.23
p-value of test : 0.77

CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

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Normal statistic          :   -0.21
p-value of test          :    0.58
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CPU time used            :   00:00:00.01
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Generator state:
 269499904 bits have been read.
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HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin
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sstring_AutoCor test:
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-----
      N =  1,   n = 16777154,   r =  0,   s = 32,   d = 2
-----
```

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Normal statistic          :   -0.055
p-value of test          :    0.52
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CPU time used            :   00:00:00.02
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Generator state:
 286277088 bits have been read.
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*****
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HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin
```

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sstring_Run test:
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      N =  1,   n = 3355443,   r =  0,   s =  32
-----
```

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Total number of 1 runs:  3355450
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Number of degrees of freedom      :   36
Chi2 statistic for number of runs :   30.57
p-value of test                   :    0.72
-----
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```
-----
Total number of bits:  13422720
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```

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Normal statistic for number of bits :    0.18
p-value of test                   :    0.43
```

CPU time used : 00:00:00.05

Generator state:
299699808 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 3.29
p-value of test : 0.35

CPU time used : 00:00:00.06

Generator state:
316477024 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 3.09
p-value of test : 0.21

CPU time used : 00:00:00.08

Generator state:
333168224 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42
ChiSquare statistic	:	46.64
p-value of test	:	0.29

Test on the values of the Statistic M

Number of degrees of freedom	:	42
ChiSquare statistic	:	33.35
p-value of test	:	0.83

Test on the values of the Statistic J

Number of degrees of freedom	:	64
ChiSquare statistic	:	52.95
p-value of test	:	0.84

Test on the values of the Statistic R

Number of degrees of freedom	:	36
ChiSquare statistic	:	20.76
p-value of test	:	0.98

Test on the values of the Statistic C

Number of degrees of freedom	:	22
ChiSquare statistic	:	17.77
p-value of test	:	0.72

CPU time used : 00:00:00.05

Generator state:
349945440 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94
ChiSquare statistic	:	84.88
p-value of test	:	0.74

Test on the values of the Statistic M

Number of degrees of freedom	:	94
ChiSquare statistic	:	71.48
p-value of test	:	0.96

Test on the values of the Statistic J

Number of degrees of freedom	:	512
ChiSquare statistic	:	503.29
p-value of test	:	0.60

Test on the values of the Statistic R

Number of degrees of freedom	:	90
ChiSquare statistic	:	83.53
p-value of test	:	0.67

Test on the values of the Statistic C

Number of degrees of freedom	:	50
ChiSquare statistic	:	75.69
p-value of test	:	0.01

CPU time used : 00:00:00.05

Generator state:

366722656 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config19_seed1.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	133.28
p-value of test	:	0.40

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	128.41
p-value of test	:	0.55

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	197.48
p-value of test	:	0.03

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	123.53
p-value of test	:	0.64

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	74.28
p-value of test	:	0.87

CPU time used : 00:00:00.02

Generator state:

383499456 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:06.08

All tests were passed